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Non-combustible aerosol system and pre-aerosol formulation housing

Abstract

A non-combustible aerosol system includes a heater configured to supply heat to a heating chamber and a housing configured to be inserted into the heating chamber. The housing defines an internal volume for containing a solid substrate. The housing includes a plurality of internal structures extending from a first end of the housing to a second end of the housing, the plurality of internal structures extending through the internal volume, the plurality of internal structures being configured to heat the solid substrate to generate an aerosol by conducting the heat supplied by the heater to the internal volume.

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Background/Summary

PRIORITY (1) This non-provisional patent application is a continuation of U.S. application Ser. No. 16/251,452, filed on Jan. 18, 2019, the entire contents of each of which are incorporated herein by reference.

BACKGROUND

Field

(1) At least some example embodiments relate generally to a non-combustible aerosol device.

Description of Related Art

(2) A non-combustible aerosol device may have a heater that heats a solid substrate, such as tobacco, without causing combustion of the solid substrate.

(3) The non-combustible aerosol device includes a power supply, such as a rechargeable battery,

arranged in the device. The battery is electrically connected to the heater, such that the heater heats the solid substrate.

SUMMARY

(4) Some example embodiments provide a non-combustible aerosol system including a heater configured to supply heat to a heating chamber and a housing configured to be inserted into the heating chamber, the housing defining an internal volume for containing a solid substrate.

(5) In some example embodiments, the solid substrate is a tobacco material which may include material from any member of the genus *Nicotiana*. In some example embodiments, the tobacco material includes a blend of two or more different tobacco varieties. Examples of suitable types of tobacco materials that may be used include, but are not limited to, flue-cured tobacco, Burley tobacco, Dark tobacco, Maryland tobacco, Oriental tobacco, rare tobacco, specialty tobacco, blends thereof and the like. The tobacco material may be provided in any suitable form, including, but not limited to, tobacco lamina, processed tobacco materials, such as volume expanded or puffed tobacco, processed tobacco stems, such as cut-rolled or cut-puffed stems, reconstituted tobacco materials, blends thereof, and the like. In some example embodiments, the tobacco material is in the form of a substantially dry tobacco mass.

(6) In some example embodiments, the tobacco material is mixed and/or combined with at least one of propylene glycol, glycerin or sub-combinations thereof or combinations thereof.

(7) The housing includes a plurality of internal structures extending from a first end of the housing to a second end of the housing, the plurality of internal structures extending through the internal volume, the plurality of internal structures being configured to heat the solid substrate to generate an aerosol by conducting the heat supplied by the heater to the internal volume.

(8) In some example embodiments, the housing further includes a first plate defining the first end of the housing and a second plate defining the second end of the housing, the first and second ends of the housing being opposite sides of the housing.

(9) In some example embodiments, the first plate and the second plate are a first material and the plurality of internal structures are a second material, the first material and the second material are different.

(10) In some example embodiments, the first plate, the second plate and the plurality of internal structures are a same material.

(11) In some example embodiments, the housing further includes an outer wall defining the internal volume, the outer wall being one of solid and a mesh.

(12) In some example embodiments, the outer wall is solid and the non-combustible aerosol system includes a piercing element configured to pierce the housing and generate an outlet for the aerosol.

(13) In some example embodiments, the outer wall is a mesh and provides an outlet for the aerosol.

(14) In some example embodiments, the housing further includes a first plate defining the first end of the housing and a second plate defining the second end of the housing, the first and second ends of the housing being opposite sides of the housing, the outer wall extending from the first plate to the second plate.

(15) In some example embodiments, the housing is cylindrical.

(16) In some example embodiments, the non-combustible aerosol system is not configured to supply an electrical current to the housing.

(17) In some example embodiments, the solid substrate includes at least one of tobacco leaf, reconstituted tobacco, compressed tobacco rod, powdered tobacco, a sub-combination thereof or a combination thereof.

(18) In some example embodiments, the non-combustible further includes a first outlet on a first side of the aerosol forming device and a second outlet on a second side of the aerosol forming device.

(19) In some example embodiments, the second outlet is a one-way valve.

(20) In some example embodiments, the plurality of internal structures are different materials.

- (21) In some example embodiments, the plurality of internal structures extend in a direction that traverses a longitudinal axis of the aerosol forming device.
- (22) In some example embodiments, the non-combustible further includes a first plate defining the first end of the housing and a second plate defining the second end of the housing, the first and second ends of the housing being opposite sides of the housing, the plurality of internal structures extending from the first plate to the second plate.
- (23) At least one example embodiment includes a method of operating a non-combustible aerosol system. The method includes inserting a housing into a non-combustible aerosol device, the housing defining an internal volume for containing a solid substrate, the housing including a plurality of internal structures extending from a first end of the housing to a second end of the housing, the plurality of internal structures extending through the internal volume, the plurality of internal structures being configured to heat the solid substrate to generate an aerosol by conducting the heat supplied by the heater to the internal volume, and activating the non-combustible aerosol device.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The various features and advantages of the non-limiting embodiments herein may become more apparent upon review of the detailed description in conjunction with the accompanying drawings. The accompanying drawings are merely provided for illustrative purposes and should not be interpreted to limit the scope of the claims. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. For purposes of clarity, various dimensions of the drawings may have been exaggerated.
- (2) FIG. 1 is a side view of a non-combustible aerosol system according to some example embodiments;
- (3) FIG. 2 illustrates a pre-aerosol formulation housing according to some example embodiments;
- (4) FIG. 3 illustrates a cross-sectional view of the non-combustible aerosol system of FIG. 2;
- (5) FIG. 4 illustrates a pre-aerosol formulation housing according to some example embodiments;
- (6) FIG. 5 illustrates a side view of a non-combustible aerosol system using the pre-aerosol formulation housing of FIG. 4 according to some example embodiments;
- (7) FIG. 6 illustrates a cross-sectional view of the non-combustible aerosol system of FIG. 5;
- (8) FIGS. 7A-7C illustrate some example embodiments of a pre-aerosol formulation housing;
- (9) FIG. 8 is a side view of non-combustible aerosol system according to some example embodiments;
- (10) FIG. 9 illustrates a cross-sectional view of the non-combustible aerosol device of FIG. 8; and
- (11) FIG. 10 illustrates a method of operating a non-combustible aerosol system according to some example embodiments.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

- (12) Some detailed example embodiments are disclosed herein. However, specific structural and functional details disclosed herein are merely representative for purposes of describing example embodiments. Example embodiments may, however, be embodied in many alternate forms and should not be construed as limited to only the example embodiments set forth herein.
- (13) Accordingly, while example embodiments are capable of various modifications and alternative forms, example embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit example embodiments to the particular forms disclosed, but to the contrary, example embodiments are to cover all modifications, equivalents, and alternatives thereof. Like numbers refer to like elements throughout the description of the figures.

(14) It should be understood that when an element or layer is referred to as being “on,” “connected to,” “coupled to,” “attached to,” “adjacent to,” or “covering” another element or layer, it may be directly on, connected to, coupled to, attached to, adjacent to or covering the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout the specification. As used herein, the term “and/or” includes any and all combinations or sub-combinations of one or more of the associated listed items.

(15) It should be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, or section from another region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of example embodiments.

(16) Spatially relative terms (e.g., “beneath,” “below,” “lower,” “above,” “upper,” and the like) may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It should be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(17) The terminology used herein is for the purpose of describing various example embodiments only and is not intended to be limiting of example embodiments. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(18) When the words “about” and “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value include a tolerance of $\pm 10\%$ around the stated numerical value, unless otherwise explicitly defined.

(19) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, including those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(20) Hardware may be implemented using processing or control circuitry such as, but not limited to, one or more processors, one or more Central Processing Units (CPUs), one or more microcontrollers, one or more arithmetic logic units (ALUs), one or more digital signal processors (DSPs), one or more microcomputers, one or more field programmable gate arrays (FPGAs), one or more System-on-Chips (SoCs), one or more programmable logic units (PLUs), one or more microprocessors, one or more Application Specific Integrated Circuits (ASICs), or any other device or devices capable of responding to and executing instructions in a defined manner.

(21) Aerosol, vapor and dispersion are terms used interchangeably and are meant to cover any

matter generated or output by the devices claimed and equivalents thereof. The pre-aerosol formulation may also be a pre-vapor formulation or a pre-dispersion formulation.

(22) FIG. 1 is a side view of non-combustible aerosol system according to at least one example embodiment. As shown in FIG. 1, a non-combustible aerosol system **10** includes a non-combustible aerosol device **100** and a pre-aerosol formulation housing **200**. The pre-aerosol formulation housing **200** may include pre-aerosol formulation that is a solid substrate (in example embodiments with tobacco, referred to as a tobacco housing).

(23) The non-combustible aerosol device **100** may include a power section **105** and a heating section **110**. In FIG. 1, the non-combustible aerosol device **100** includes a housing **115**. In at least one example embodiment, the housing **115** may have a generally square cross-section. In other example embodiments, the housing **115** may have a generally triangular or circular cross-section.

(24) As shown, a power section **105** and a heating section **110** are integral sections of the housing **115**. However, it should be understood that example embodiments are not limited thereto and the power section **105** and heating section **110** may have separate detachable housings. For example, the heating section **110** may be a replaceable cartridge and the power section **105** may be a reusable battery section. The power section **105** and the heating section **110** may be coupled together by any type of connector, such as a snug-fit, detent, clamp, bayonet, and/or clasp.

(25) The housing **115** extends in a longitudinal direction between a first end **130** and a second end **132**.

(26) At least one air inlet **120** extends through a portion of the housing **115**. In at least one example embodiment, the at least one air inlet **120** may be formed in the housing **115** to control a resistance-to-draw (RTD) during use. In at least one example embodiment, the air inlet **120** may be machined into the housing **115** with precision tooling such that their diameters are closely controlled and replicated from one non-combustible aerosol device **100** to the next during manufacture.

(27) In at least one example embodiment, the air inlet **120** may be sized and configured such that the non-combustible aerosol device **100** has a desired resistance-to-draw (RTD) range of 20 to 150 mm of water.

(28) The non-combustible aerosol device **100** includes a mouthpiece **125** at the first end **130** of the non-combustible aerosol device **100**. As shown in FIG. 1, the heating section **110** is at a proximal end **135** of the non-combustible aerosol device **100** (relative to the mouthpiece **125**) and the power section **105** is at a distal end **134** of the non-combustible aerosol device **100** (relative to the mouthpiece **125**).

(29) At least one side of the housing **115** in the heating section **110** defines an opening **140** of a channel space within the housing **115** in the non-combustible aerosol device **100**. In some example embodiments, the opening **140** may have a same shape as the pre-aerosol formulation housing **200**. While the opening **140** is illustrated as being elliptical, example embodiments are not limited thereto. For example, the opening **140** may be circular, rectangular, triangular or another polygon.

(30) The opening **140** may be configured to receive, into a channel space of the non-combustible aerosol device **100**, the pre-aerosol formulation housing **200**.

(31) FIG. 2 illustrates an example embodiment of the pre-aerosol formulation housing **200**. A cross-sectional shape of the pre-aerosol formulation housing **200** may have a same shape as the opening **140** and may have an outer diameter that corresponds to a diameter **142** of the opening **140**. As shown in FIG. 2, the pre-aerosol formulation housing **200** is an encased capsule and includes a first plate **205**, a second plate **210** and a lateral outer wall **215**. The outer wall **215**, the first plate **205** and the second plate **210** form the encased capsule. The pre-aerosol formulation housing **200** may be cylindrical with the first plate **205** and the second plate **210** being circular ends, respectively, of the pre-aerosol formulation housing **200**. The wall **215** extends from the first plate **205** to the second plate **210** and defines an internal volume **V** that has a diameter corresponding to diameters (e.g., the same) of the first plate **205** and the second plate **210**. The outer wall **215** closes off the internal volume **V** between the first plate **205** and the second plate **210**

to prevent the pre-aerosol formulation from escaping the pre-aerosol formulation housing **200**.

(32) A pliable material such as a polymer can be used on the plates **205** and **210** of the pre-aerosol formulation housing **200** and on the opening **140** which creates a seal when the pre-aerosol formulation housing **200** is inserted into the housing **115**.

(33) A plurality of internal structures **220** extend from the first plate **205** to the second plate **210**. In some example embodiments, the internal structures **220** are elongated and have a longitudinal direction traversing an air flow direction when inserted in the housing **115**. In the example embodiments shown in FIG. 2, the internal structures **220** have a square cross section. However, the internal structures **220** may be any cross-sectional shape such as rectangular, oval and circular. Moreover, each of the internal structures **220** may have a different cross sectional shape.

(34) The internal structures **220** are made of a material that conducts heat to heat a pre-aerosol formulation within the pre-aerosol formulation housing **200** without combustion occurring. For example, a heater (e.g., shown in FIG. 3) and the pre-aerosol formulation housing **200** are configured to heat the pre-aerosol formulation to a temperature ranging from 100 to 350 degrees Celsius to produce an aerosol. The internal structures **220** create more surface area contact with the pre-aerosol formulation and improve a heating efficiency of the pre-aerosol formulation.

(35) The pre-aerosol formulation housing **200** may be made of any material that conducts heat. In some example embodiments, the pre-aerosol formulation housing **200** may be made of metal and, thus, may be referred to as a metallic housing. The internal structures **220** may be made of the same material as the first plate **205**, the second plate **210** and the outer wall **215** or may be made of a different material. Moreover, the internal structures **220** may all be made of the same material or at least one of the internal structures **220** may be made of a different material than the remaining internal structures.

(36) The internal structures **220** are attached to the first plate **205** and the second plate **210** by any means that permits the internal structure **220** to sufficiently conduct heat from the plates **205**, **210** such as soldering, welding or a male/female friction fit connection.

(37) For example, the internal structures **220** may be made of pure metals, alloys, and/or polymers.

(38) A diameter **225** of the plates **205**, **210** may be 1-20 millimeters and a width **227** of the pre-aerosol formulation housing **200** may be 1-20 millimeters. In some example embodiments, the width **227** traverses the longitudinal axis of the non-combustible aerosol device **100**. The width **227** may be the same or smaller than a length of the chamber **144**.

(39) As shown in FIG. 2, the outer wall **215** is a mesh in some example embodiments, but is not limited thereto. When the outer wall **215** is a mesh, the generated aerosol may escape the pre-aerosol formulation housing **200** through holes in the mesh and flow to the mouthpiece **125**.

(40) The pre-aerosol formulation housing **200** may house a pre-aerosol formulation with the first plate **205**, the second plate **210** and the outer wall **215** being configured to conduct heat from the heater **148** to heat the pre-aerosol formulation.

(41) FIG. 3 illustrates a cross-sectional view of the non-combustible aerosol system **10** according to some example embodiments. As shown in FIG. 3, the opening **140** may be configured to receive, into a channel space of the non-combustible aerosol device **100**, the pre-aerosol formulation housing **200**. The pre-aerosol formulation housing **200** may have a same shape as the opening **140** and may have an outer diameter that corresponds to the diameter **142** of the opening **140**. In some example embodiments, the pre-aerosol formulation housing **200** has an outer diameter smaller than the diameter **142** such that the pre-aerosol formulation housing **200** may be inserted into the opening **140**. While the pre-aerosol formulation housing **200** is illustrated as having an elliptical cross-section (cut normal to a longitudinal axis of the pre-aerosol formulation housing **200**), example embodiments are not limited thereto. For example, the pre-aerosol formulation housing **200** may have a cross section that is circular, rectangular, triangular or another polygon.

(42) The opening **140** provides access to a chamber **144**. In some example embodiments, the chamber **144** has a same shape as the opening **140**. A diameter of the chamber **144** may be the same

as the diameter **142**. A seal is formed between the housing **115** and the pre-aerosol formulation housing **200** due to the specific size of the pre-aerosol formulation housing **200** and the chamber **144** providing a secure fit. A seal can also be formed by spring loaded plates (not shown) at the front and back of the chamber **144** that provide pressure on the front and back of the pre-aerosol formulation housing **200** once the pre-aerosol formulation housing **200** is inserted.

(43) In some example embodiments, ends of the pre-aerosol formulation housing **200** may be exposed to ambient air once inserted into the non-combustible aerosol device **100**. Alternatively, the non-combustible aerosol device **100** may include a cover that covers the opening **140** once the pre-aerosol formulation housing **200** is inserted into the non-combustible aerosol device **100**. The cover may be hinged, slidable, spring loaded or another type of cover.

(44) In some example embodiments, the housing **115** may include an opening on an opposite side of the side with the opening **140** that opposes the opening **140**. In other example embodiments, the opposite side of the side with the opening **140** does not have an opening.

(45) The pre-aerosol formulation housing **200** may be inserted into a channel space defined by the chamber **144**, via the opening **140**, so that the pre-aerosol formulation housing **200** is exposed to heat generated by a heater **148**. In some example embodiments, the pre-aerosol formulation housing **200** does not contact a heat source such as the heater **148** and/or an electrical source. Rather, the pre-aerosol formulation housing **200** is made of a material that conducts heat generated from the heater **148** to heat a pre-aerosol formulation **202** (e.g., tobacco) in the pre-aerosol formulation housing **200** and, thus, no electrical heating of the pre-aerosol formulation housing **200** occurs.

(46) The pre-aerosol formulation housing **200** may conduct the heat generated by the heater **148** to the pre-aerosol formulation **202** to an extent that flavoring, nicotine and/or ingredients in the pre-aerosol formulation **202** is at least partially extracted (e.g., aerosolized) to create a downstream aerosol **350b** (and a bypass airflow **355** that may contain aerosol) that is extracted from the pre-aerosol formulation **202**. The pre-aerosol formulation housing **200** heats the pre-aerosol formulation **202** to an extent that the pre-aerosol formulation **202** and the flavoring, nicotine and/or pre-aerosol formulation remain below a combustion temperature. That is to say, in some example embodiments, the pre-aerosol formulation housing **200** does not combust any material in the pre-aerosol formulation **202**, including the flavoring, nicotine and/or pre-aerosol formulation.

(47) As described, the pre-aerosol formulation housing **200** includes at least an outer wall **215** and internal structures **220** that conduct heat to heat the pre-aerosol formulation **202**. While the internal structures **220** are illustrated as traversing the longitudinal axis of the non-combustible aerosol device **100**, example embodiments are not limited thereto and the internal structures **220** may be parallel to the longitudinal axis of the non-combustible aerosol device **100**.

(48) In some example embodiments, the heater **148** contacts the pre-aerosol formulation housing **200**. In other example embodiments, the heater **148** is at a distance from the heater **148** to generate a desired temperature.

(49) The power section **105** includes a power supply **152** and a circuit board **156** arranged in the non-combustible aerosol device **100**. The power supply **152** and the circuit board **156** may be mounted on a common support **160**. The common support **160** is structured to fit within the housing **115** to eliminate/reduce movement of the power supply **152** and the circuit board **156** when the non-combustible aerosol device **100** is in use.

(50) The power supply **152** may be a Lithium-ion battery or one of its variants, for example a Lithium-ion polymer battery. Alternatively, the power supply **152** may be a nickel-metal hydride battery, a nickel cadmium battery, a lithium-manganese battery, a lithium-cobalt battery, solar cell or a fuel cell. The non-combustible aerosol device **100** may be used until the energy in the power supply **152** is depleted or in the case of lithium polymer battery, a minimum voltage cut-off level is achieved.

(51) The circuit board **156** may include at least control circuitry **162**, an air flow sensor **164** and a

memory **165**. The circuit board **156** may include other circuitry such as communications circuitry **166** (e.g., Bluetooth™) to communicate wirelessly with an external device such as a mobile phone. Thus, the non-combustible aerosol device **100** is not limited to the circuitry shown in FIG. **3**. The memory **165** (e.g., a tangible storage medium) may be read-only memory (ROM), random access memory (RAM), or flash memory (e.g., USB flash drives, memory cards, memory sticks, etc.), for example. Example embodiments are not limited by these aspects of any given implementation.

(52) In at least one example embodiment, the control circuitry **162** may include at least one processor. In this example, the processor may be any known, or to be developed, processor configured to execute computer-readable instructions stored on the memory **165**. Execution of the computer-readable instructions stored on the memory **165** transforms the at least one processor into a special purpose processor for carrying out the functionality described herein. The memory **165** may be further configured to store various types of information regarding the non-combustible aerosol system **10**, such as that described above.

(53) Although discussed in some cases with regard to a processor and a memory, according to at least some example embodiments, the control circuitry **162** (or control circuitry or processing circuitry) may be (or include) hardware, firmware, hardware executing software, or any combination thereof. For example, the control circuitry **162** may include one or more Central Processing Units (CPUs), digital signal processors (DSPs), application-specific-integrated-circuits (ASICs), field programmable gate arrays (FPGAs), or other circuitry configured as special purpose machines to perform the functions of the control circuitry **162**.

(54) The control circuitry **162** is connected to the power supply **152** by a cathode connector **170a** and an anode connector **170b**. The control circuitry **162** is configured to supply power to electrical components (e.g., the heater **148**, the air flow sensor **164** and the memory **165**) of the non-combustible aerosol device **100** using power supplied from the power supply **152**.

(55) The power section **105** may be separated from the heating section **110** by a divider **172**. The divider **172** may be a gasket (or seal) that provides a substantially tight seal with an interior surface of the housing **115**. The divider **172** may include a channel **174** disposed between the power section **105** and the heating section **110** to allow a negative pressure to be applied in the power section **105** when air is pulled through the air inlet **120** (e.g., when air is pulled through an outlet **176** of the mouthpiece **125**).

(56) The divider **172** may also include holes **178a** and **178b**. The holes **178a** and **178b** are sized to fit wires **180a** and **180b**. In other example embodiments, electrical contacts may be used instead of the wires **180a** and **180b**. The holes **178a** and **178b** may be sized so no air flows between the power section **105** and the heating section **110** through the holes **178a** and **178b**. In other example embodiments, seals may be in the holes **178a** and **178b** to prevent air flowing between the power section **105** and the heating section **110**.

(57) The heater **148** may extend transversely across an inner passage **182** between opposing walls of the housing **115**. In some example embodiments, the heater **148** may extend parallel to a longitudinal axis of the inner passage **182**.

(58) The power section **105** may further include an end cap **151** at the second end **132**. The end cap **151** may seal off the second end **132**. The end cap **151** may attach to the housing **115** using known connection systems such as threaded connectors and/or friction fit connection systems. The end cap **151** may include electrical contacts **153a** and **153b** for charging the power supply **152**.

(59) In another example embodiment, the end cap **151** may be an integral portion of the housing **115**.

(60) In at least one example embodiment, the power supply **152** is rechargeable. The power section **105** may include circuitry configured to allow the battery to be chargeable by an external charging device. To recharge the non-combustible aerosol device **100**, an USB charger or other suitable charger assembly may be used by connecting an external charger to the electrical contacts **153a** and **153b**. The control circuitry **162** is connected to the electrical contacts **153a** and **153b** and controls

the supply of power to the power supply **152**.

(61) The heater **148** can be in the form of a wire coil, a planar body, a ceramic body, a single wire, a cage of resistive wire or any other suitable form. The heater **148** may be any heater that is configured to heat a pre-aerosol formulation to a sufficient temperature to generate an aerosol.

(62) In at least one example embodiment, the heater may be formed of any suitable electrically resistive materials. For example, the heater **148** may include at least one material selected from the group consisting of stainless steel, copper, copper alloys, nickel-chromium alloys, super alloys, or sub-combinations thereof or combinations thereof. In an example embodiment, the heater **148** may be formed of nickel-chromium alloys or iron-chromium alloys. In another example embodiment, the heater **148** may be a ceramic heater having an electrically resistive layer on an outside surface thereof.

(63) In at least one example embodiment, the heater **148** may heat the pre-aerosol formulation housing **200** by thermal conduction. Alternatively, heat from the heater **148** may be conducted to the pre-aerosol formulation housing **200** by means of a heat conductive element or the heater **148** may transfer heat to the incoming ambient air that is drawn through the non-combustible aerosol device **100** during use, which in turn heats the pre-aerosol formulation housing **200** by convection.

(64) The pre-aerosol formulation housing **200** resides in or near an airflow path **350a** that is defined by the non-combustible aerosol device **100**. This airflow path **350a** may be formed, for instance, by the air inlets **120** and the outlet **176**. The airflow path **350a** may pass across the pre-aerosol formulation housing **200** or directly through the pre-aerosol formulation housing **200**. It should be understood that, in the event the non-combustible aerosol device **100** includes a bypass airflow **355**, this bypass airflow **355** may include an entrained aerosol just as the downstream aerosol **350b** (that passed through the matrix pre-aerosol formulation housing **200**) also includes an aerosol, if the bypass airflow **355** passes across an exposed surface of the pre-aerosol formulation housing **200**.

(65) In some example embodiments, the pre-aerosol formulation includes a botanical material. For example, a botanical material may include a tobacco material.

(66) In some example embodiments, the tobacco material may include material from any member of the genus *Nicotiana*. In some example embodiments, the tobacco material includes a blend of two or more different tobacco varieties. Examples of suitable types of tobacco materials that may be used include, but are not limited to, flue-cured tobacco, Burley tobacco, Dark tobacco, Maryland tobacco, Oriental tobacco, rare tobacco, specialty tobacco, blends thereof and the like. The tobacco material may be provided in any suitable form, including, but not limited to, tobacco lamina, processed tobacco materials, such as volume expanded or puffed tobacco, processed tobacco stems, such as cut-rolled or cut-puffed stems, reconstituted tobacco materials, blends thereof, and the like. In some example embodiments, the tobacco material is in the form of a substantially dry tobacco mass.

(67) In some example embodiments, the tobacco material is mixed and/or combined with at least one of propylene glycol, glycerin or sub-combinations thereof or combinations thereof.

(68) In an example embodiment, flavoring, a flavorant, or a flavor system, is in the pre-aerosol formulation in order to release an aroma and/or flavors during operation, including in some cases, upon heating and/or as an airflow passes through the non-combustible aerosol device **100**. In an example embodiment, the flavoring includes volatile tobacco flavor compounds. Flavoring may also include flavors besides tobacco, or in addition to tobacco flavoring. The flavoring may be at least one flavorant that is a natural flavorant or an artificial flavorant. For instance, the at least one flavorant may include tobacco flavor, tobacco extract, menthol, wintergreen, peppermint, herb flavors, fruit flavors, nut flavors, liquor flavors, roasted, minty, savory, cinnamon, clove, and any other desired flavors, and combinations or sub-combinations thereof. In some example embodiments, a pre-aerosol formulation housing **200** that includes a tobacco material is referred to as a tobacco element.

(69) In operation, with the non-combustible aerosol device **100** in an assembled configuration, a negative pressure may be applied on the mouthpiece **125**. For example, the negative pressure may be drawn upon the mouthpiece **125**. This negative pressure may cause an internal pressure drop inside the non-combustible aerosol device **100** that may cause an inlet air flow to enter the non-combustible aerosol device **100** via the air inlets **120**. The internal pressure drop may also cause an internal pressure drop within the heating section **110** as air is drawn through air inlet **120** (via an air flow path traveling through section **110**).

(70) The air flow sensor **164** may be exposed to the channel **174**. In an example embodiment, the air flow sensor **164** generates an output signal indicative of a magnitude and direction of the airflow **350a** through the inner passage **182**, where the control circuitry **162** receives the air flow sensor **164** output signal and determine if the following internal conditions exist: (1) a direction of the airflow **350a** indicates a draw on the mouthpiece **125** (versus blowing air through the mouthpiece **125**), and/or (2) a magnitude of the airflow **350a** exceeds a threshold value. In some example embodiments, only one condition may be sufficient to activate the heater **148**, while in other examples, two conditions or all conditions may have to be met before activating the heater. If these internal conditions of the non-combustible aerosol device **100** are met, the control circuitry **162** electrically closes the electrical circuit to connect the power supply **152** to the heater **148**, thereby activating the heater **148** by sending an electrical current to the heater **148**. In an example embodiment, the air flow sensor **164** generates a variable output signal that is in at least partial correlation with a magnitude of the pressure drop sensed by the sensor **106**.

(71) The air flow sensor **164** may be a sensor as disclosed in “Electronic Smoke Apparatus,” U.S. application Ser. No. 14/793,453, filed on Jul. 7, 2015, or a sensor as disclosed in “Electronic Smoke,” U.S. Pat. No. 9,072,321, issued on Jul. 7, 2015, each of which are hereby incorporated by reference in their entirety into this document. Other type of sensors to detect an airflow may be used.

(72) In at least some example embodiments, the non-combustible aerosol device **100** may include a temperature sensor to monitor a temperature of the heater **148** and feedback the sensed temperature to the control circuitry **162**. The control circuitry **162** may use the sensed temperature and/or the sensed pressure drop to control the power supplied to the heater **148**.

(73) Wires **180a** and **180b** carry an electrical current to the heater **148** in order to energize the heater **148**. The energized heater **148** in turn heats the pre-aerosol formulation housing **200**. The first plate **205**, the second plate **210**, the outer wall **215** and the internal structures **220** conduct the heat from the heater **148** to heat the flavor material within the pre-aerosol formulation housing **200**. The pre-aerosol formulation housing **200** conducts the heat from the heater **148** to heat the flavor material to a temperature sufficient to generate an aerosol without combusting the flavor material (e.g., 100 to 300 degrees Celsius).

(74) In another example embodiment, the non-combustible aerosol device **100** may include a push button to operate the non-combustible aerosol device **100** and cause the power supply **152** and the control circuitry **162** to supply power to the heater **148**.

(75) The aerosol may elute nicotine and/or tobacco elements into the flow stream. Some thermal reactions may also be present between the aerosol and the tobacco element.

(76) FIG. 4 illustrates a pre-aerosol formulation housing according to some example embodiments. As shown in FIG. 4, a pre-aerosol formulation housing **200a** includes an outer wall **215a** that is a solid material.

(77) In example embodiments including the pre-aerosol formulation housing **200a**, the non-combustible aerosol device **100** includes a piercing mechanism to pierce at least one of the outer wall **215a**, the first plate **205** and the second plate **210** upon the pre-aerosol formulation housing **200a** being inserted into the non-combustible aerosol device **100**. The piercing generates an outlet for the aerosol that is generated with a flavor material within the pre-aerosol formulation housing **200a** when the pre-aerosol formulation housing **200** is heated to an aerosol producing temperature

(e.g., 100 to 350 degrees Celsius).

(78) FIG. 5 illustrates a side view of non-combustible aerosol system using the pre-aerosol formulation housing of FIG. 4 according to some example embodiments.

(79) FIG. 6 illustrates a cross-sectional view of the non-combustible aerosol system of FIG. 5.

(80) A non-combustible aerosol system **10a**, shown in FIG. 5, is the same as the non-combustible aerosol system **10** except a non-combustible aerosol device **100a** includes a button **605** at the first end **130** to actuate a piercing structure.

(81) As shown in FIG. 6, the button **605** actuates a piercing structure **610** to pierce the pre-aerosol formulation housing **200a**. In an example embodiment, the piercing structure **610** includes a first piercing element **615** (e.g., a blade), a second piercing element **620** (e.g., a blade), rods **625a-625c**, a first link **630** and a second link **635**.

(82) The rod **625a** extends in the longitudinal axis of the non-combustible aerosol device **100a** and couples the button **605** and the first piercing element **615**. The first link **630** couples the rod **625a** to a first end **625b1** of the rod **625b**. The first link **630** may be a pin that extends through the rod **625a** and the rod **625b** to permit the rod **625b** to rotate along an axis of the first link **630**. The axis of the first link **630** about which the rod **625b** may rotate may be transverse to the longitudinal axis of the non-combustible aerosol device **100a**. In other example embodiments, the first link **630** may maintain the positional relationship between the rod **625a** and the rod **625b**.

(83) The rod **625b** extends over a chamber **144a** from the first end **625b1** to a second end **625b2**. The chamber **144a** may differ from the chamber **144** in that the chamber **144a** includes two semicircular sections **632a** and **632b**. In at least a portion of the chamber **144a** along the longitudinal axis of the chamber **144a**, the semicircular sections **632a** and **632b** define openings **635a** and **635b** between the semicircular sections **632a** and **632b**.

(84) Similar to the rod **625b**, the rod **625c** extends over the chamber **144a** from a first end **625c1** to a second end **625c2**.

(85) The second link **635** couples the second end **625b2** of the rod **625b** to the second end **625c2** of the rod **625c**. The second link **635** may be a pin (e.g., spring loaded) that allows both the rod **625b** to rotate about the second link **635**. As shown in the example embodiment of FIG. 6, the second piercing element **620** is integral with the rod **625c**. The first end **625c1** is angled toward the pre-aerosol formulation housing **200a** such that the second piercing element **620** faces the pre-aerosol formulation housing **200a**. However, it should be understood example embodiments are not limited thereto.

(86) When an adult aerosol consumer pushes the button **605** toward the inner passage **182**, the force applied by the adult aerosol consumer causes the rod **625a** and, more specifically, the first piercing element **615** to travel towards the pre-aerosol formulation housing **200a** and pierce the wall **215a** of the pre-aerosol formulation housing **200a**. The second link **635** permits the rod **625b** to rotate in a direction **640**. The force applied by the first piercing element **615** onto the pre-aerosol formulation housing **200a** causes the pre-aerosol formulation housing **200a** to move in a direction the same as the longitudinal axis of the non-combustible aerosol device **100a** or a substantially similar direction. The pre-aerosol formulation housing **200a** moves to the second piercing element **620**, which pierces the pre-aerosol formulation housing **200a** when the adult aerosol consumer applies the force to the button **605**. The piercing by the first piercing element **615** and the second piercing element **620** create an airflow path **650** through the pre-aerosol formulation housing **200a** to the outlet **176**.

(87) While FIG. 6 illustrates a button activated piercing structure, example embodiments are not limited thereto. For example, a piercing structure using a lever and release spring to pierce a pre-aerosol formulation housing may be used.

(88) FIG. 7A illustrates an example embodiment of a pre-aerosol formulation housing. As shown in FIG. 7A, a pre-aerosol formulation housing **200b** is the same as the pre-aerosol formulation housing **200** except plates **205a** and **210** are a mesh instead of solid. The mesh may be a screen-

type of material made of pure metals, metal-alloy, or polymer. The hole size may range from 0.001 to 0.1 inch. In FIG. 7A, the wall **215** is omitted for clarity.

(89) FIG. 7B illustrates an example embodiment of a pre-aerosol formulation housing. As shown in FIG. 7B, a pre-aerosol formulation housing **200c** is the same as the pre-aerosol formulation housing **200** except the internal structures **220a** have a circular cross-section instead of a square cross-section. The wall **215** is omitted for clarity.

(90) FIG. 7C illustrates an example embodiment of a pre-aerosol formulation housing. As shown in FIG. 7C, a pre-aerosol formulation housing **200d** is the same as the pre-aerosol formulation housing **200** except plates **205a** and **210** are square instead of circular. The wall **215** is omitted for clarity. The wall **215** extends around each of the four edges of each plate **205c** and **210c**. A height **225c** of the plates **205c** and **210c** may be between 1-20 millimeters.

(91) FIG. 8 is a side view of non-combustible aerosol system according to at least another example embodiment. FIG. 9 illustrates a cross-sectional view of the non-combustible aerosol device of FIG. 8.

(92) A non-combustible aerosol system **10b**, shown in FIG. 8, is the same as the non-combustible aerosol system **10** except a heating section **110a** of a non-combustible aerosol device **100b** includes an olfactory port **305**.

(93) As shown in FIG. 9, the olfactory port **305** is located over the chamber **144** and the pre-aerosol formulation housing **200** (when the pre-aerosol formulation housing **200** is inserted into the chamber **144**). The olfactory port **305** extends at an angle α from the heating section **110**. The angle α may be greater than zero degrees and less than or equal to 90 degrees such that the adult aerosol consumer can smell the aerosol produced by the non-combustible aerosol device **100b**. The olfactory port **305** allows a portion of the air exposed to the flavorant in the pre-aerosol formulation housing **200** to flow out of the olfactory port **305** as indicated by airflow **310**.

(94) In an example embodiment, the olfactory port **305** is integral with a housing **115a** and made of the same material as the housing **115a**.

(95) In an example embodiment, the olfactory port **305** is a one way valve as to not affect the RTD. Moreover, the olfactory port **305** may be opened and closed as desired by the adult consumer. For example, a user interface such as a needle valve or slide could be attached to the housing **115a** to enable control of the olfactory port **305** by the adult consumer.

(96) FIG. 10 illustrates a method of operating a non-combustible aerosol system according to at least one example embodiment. The non-combustible aerosol system may a system encompassed described in any of FIGS. 1-9.

(97) At **S1005**, an adult aerosol consumer inserts a pre-aerosol formulation housing into a non-combustible aerosol device. The pre-aerosol formulation housing defines an internal volume for containing a pre-aerosol formulation and includes a plurality of internal structures extending from a first end of the housing to a second end of the housing. The plurality of internal structures extend through the internal volume and are configured to heat the pre-aerosol formulation to generate an aerosol by conducting the heat supplied by a heater to the internal volume. For example, the adult aerosol consumer may insert the pre-aerosol formulation housing **200** into the non-combustible aerosol device **100**.

(98) At **S1010**, the adult aerosol consumer activates the non-combustible aerosol device by applying a negative pressure at a mouthpiece. The negative pressure causes a sensor to send to control circuitry a signal representing the pressure in the non-combustible aerosol device. Based on the signal, the control circuitry activates the heater. The pre-aerosol formulation housing may conduct the heat generated by the heater to the pre-aerosol formulation to an extent that the flavoring, nicotine and/or ingredients in the pre-aerosol formulation is at least partially extracted (e.g., aerosolized) to create a downstream aerosol that is extracted from the pre-aerosol formulation.

(99) Example embodiments have been disclosed herein, it should be understood that other

variations may be possible. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

Claims

1. A cartridge for a non-combustible aerosol system, comprising: a cartridge housing defining an opening to a chamber in at least one side of the cartridge housing; a heater configured to supply heat to the chamber in the cartridge housing, the heater extending transversely across an inner passage of the cartridge housing; and a pre-aerosol formulation housing configured to be inserted into the chamber, the pre-aerosol formulation housing including, a first plate, a second plate opposite the first plate, an outer wall extending between the first plate and the second plate, and a plurality of internal structures extending from the first plate to the second plate, the plurality of internal structures configured to conduct heat supplied by the heater and heat a solid substrate within the pre-aerosol formulation housing.
2. The cartridge of claim 1, wherein the first plate and the second plate are a first material and the plurality of internal structures are a second material, the first material and the second material are different.
3. The cartridge of claim 1, wherein the first plate, the second plate and the plurality of internal structures are a same material.
4. The cartridge of claim 1, wherein the outer wall includes one of a solid and a mesh.
5. The cartridge of claim 4, wherein the outer wall is solid and the cartridge housing includes, a piercing element configured to pierce the pre-aerosol formulation housing and generate an outlet for an aerosol.
6. The cartridge of claim 4, wherein the outer wall is a mesh and provides an outlet for an aerosol.
7. The cartridge of claim 1, wherein the pre-aerosol formulation housing is cylindrical.
8. The cartridge of claim 1, wherein the first plate and the second plate include of a square or rectangular shape.
9. The cartridge of claim 1, wherein the non-combustible aerosol system is not configured to supply an electrical current to the pre-aerosol formulation housing.
10. The cartridge of claim 1, wherein the solid substrate includes at least one of tobacco leaf, reconstituted tobacco, compressed tobacco rod, powdered tobacco, a sub-combination thereof or a combination thereof.
11. The cartridge of claim 1, wherein the plurality of internal structures are different materials.
12. The cartridge of claim 1, wherein the plurality of internal structures extend in a direction that traverses a longitudinal axis of the cartridge housing.
13. The cartridge of claim 1, wherein the plurality of internal structures includes a circular cross section.
14. The cartridge of claim 1, wherein the plurality of internal structures includes a square cross section.
15. The cartridge of claim 1, wherein the cartridge housing is configured to be removably coupled to a power section housing.
16. The cartridge of claim 15, wherein the power section housing includes: a sensor configured to detect air flow in the chamber; and a controller configured to supply power to the heater based on the detected air flow.
17. The cartridge of claim 1, further comprising: a first piercing element configured to pierce a first end of the pre-aerosol formulation housing; and a second piercing element opposite the first piercing element configured to pierce a second end of the pre-aerosol formulation housing.
18. The cartridge of claim 17, wherein the first piercing element and the second piercing element are configured to generate an air flow path through the pre-aerosol formulation housing to an outlet

in the cartridge housing.

19. The cartridge of claim 17, wherein a first end of the cartridge housing includes a button configured to move the first piercing element into the first end of the pre-aerosol formulation housing and to move the pre-aerosol formulation housing toward the second piercing element in response to a force applied to the button.

20. The cartridge of claim 1, further comprising an outlet on a side of the cartridge housing.

21. The cartridge of claim 20, wherein the outlet is a one-way valve.

22. The cartridge of claim 20, wherein the outlet is an olfactory port adjacent the chamber and air exposed to a flavorant in the pre-aerosol formulation housing is configured to flow out of the olfactory port.

23. The cartridge of claim 20, wherein the outlet extends at an angle from the cartridge housing.
